

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS:5

Total Marks 50

Duration :60 Minutes Annexure A

Comparative Analysis

Code of Conduct:

I S.Jeswanth Kumar-192321056 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Jeswanth Kumar

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	

Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
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Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Depth Estimation Comparison (Stereo vs SfM) Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4

- Compute absolute error for both methods.
- Compare and evaluate which method is more accurate.

2. Pattern Recognition Accuracy

Comparison Two recognition systems

produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

- Compute accuracy for both systems.
- Compare and evaluate which system performs better.

3. Precision-Recall Trade-off Analysis

Two models produce:

Model	TP	FP	FN
A	70	30	20
B	85	25	25

- Compute precision and recall for both models.
- Compare and evaluate which model is more balanced.

4. F1-Score Comparison Two

models:

Model	Precision	Recall
A	0.8	0.6
B	0.75	0.75

- Compute F1-score for both models.
- Compare and evaluate which model is better.

5. Optical Flow Consistency Motion

values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

Answers:

1.Depth Estimation Comparison (Stereo vs SfM)

(a) Absolute Error Calculation

Absolute Error = |Estimated Value – Actual Value|

Object	Actual Depth (m)	Stereo Vision (m)	Stereo Error	SfM (m)	SfM Error
A	2.0	2.0	0.0	2.5	0.5
B	3.0	3.0	0.0	3.8	0.8
C	4.0	4.0	0.0	5.0	1.0

Mean Absolute Error (MAE):

Stereo Vision MAE = $(0.0 + 0.0 + 0.0) / 3 = 0.0$ m

Structure from Motion (SfM) MAE = $(0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 \approx 0.767$ m

(b) Comparison and Evaluation

Stereo Vision produced zero absolute error across all three objects, exactly matching the actual depth values, while SfM showed an increasing error as depth increased (0.5 m, 0.8 m, 1.0 m), giving a mean absolute error of about 0.767 m. This indicates that Stereo Vision is significantly more accurate for this dataset. SfM's error grows with distance, which is typical because scale in SfM is estimated (not directly measured via a fixed baseline as in stereo), making it more prone to scale drift at larger depths. Conclusion: Stereo Vision is the more accurate depth estimation method here.

2. Pattern Recognition Accuracy Comparison

(a) Accuracy Calculation

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Samples}) \times 100$$

System	Correct Predictions	Total Samples	Accuracy (%)
Traditional	85	100	85%
Deep Learning	92	100	92%

$$\text{Traditional System Accuracy} = (85 / 100) \times 100 = 85\%$$

$$\text{Deep Learning System Accuracy} = (92 / 100) \times 100 = 92\%$$

(b) Comparison and Evaluation

The Deep Learning system achieves 92% accuracy compared to 85% for the Traditional system, a 7 percentage-point improvement. This is consistent with the general strength of deep learning models in automatically learning hierarchical and discriminative features from visual data, rather than relying on hand-crafted features as traditional methods do. Conclusion: The Deep Learning system performs better on this pattern recognition task.

3. Precision–Recall Trade-off Analysis

(a) Precision and Recall Calculation

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad \text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Model	TP	FP	FN	Precision	Recall
A	70	30	20	$70/100 = 0.700$	$70/90 = 0.778$
B	85	25	25	$85/110 = 0.773$	$85/110 = 0.773$

(b) Comparison and Evaluation

Model A has Precision = 0.700 and Recall = 0.778, a gap of about 0.078 between the two metrics. Model B has Precision = 0.773 and Recall = 0.773, which are essentially equal. Because Model B's precision and recall are almost identical, it demonstrates a more balanced trade-off between false positives and false negatives, whereas Model A leans toward missing more positive cases relative to the false alarms it makes. Conclusion: Model B is the more balanced model, even though Model A's recall alone is slightly higher than Model B's.

4. F1-Score Comparison

(a) F1-Score Calculation

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Model	Precision	Recall	F1-Score
A	0.80	0.60	$2 \times (0.8 \times 0.6) / (0.8 + 0.6) = 0.96 / 1.4 \approx 0.686$
B	0.75	0.75	$2 \times (0.75 \times 0.75) / (0.75 + 0.75) = 1.125 / 1.5 = 0.750$

(b) Comparison and Evaluation

Model A has a higher precision (0.80) but a noticeably lower recall (0.60), yielding an F1-score of approximately 0.686. Model B has more balanced precision and recall (0.75 each), giving a higher F1-score of 0.750. Since the F1-score rewards balance between precision and recall, Model B outperforms Model A overall, despite Model A being more precise. Conclusion: Model B is the better model based on F1-score.

5. Optical Flow Consistency

(a) Average Motion Calculation

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

Average Motion (Method A) = $(4 + 5 + 6) / 3 = 15 / 3 = 5.0$

Average Motion (Method B) = $(6 + 8 + 9) / 3 = 23 / 3 \approx 7.667$

Variance / Standard Deviation (measure of consistency):

Method A deviations from mean (5.0): $-1, 0, +1 \rightarrow \text{Variance} = (1+0+1)/3 = 0.667 \rightarrow \text{Std Dev} \approx 0.816$

Method B deviations from mean (7.667): $-1.667, +0.333, +1.333 \rightarrow \text{Variance} = (2.778+0.111+1.778)/3 \approx 1.556 \rightarrow \text{Std Dev} \approx 1.247$

(b) Comparison and Evaluation

Method B records a higher average motion (7.667) than Method A (5.0), but Method A has a lower variance (0.667) and standard deviation (≈ 0.816) compared to Method B's variance (≈ 1.556) and standard deviation (≈ 1.247). A lower standard deviation means Method A's frame-to-frame motion values are more tightly clustered around the mean, indicating steadier, more consistent optical flow estimates across frames. Conclusion: Method A is the more stable/consistent method, even though Method B detects greater overall motion magnitude.